

Lesson 1: Lists Explore

45 minutes

Overview

Students will learn the ways that lists are created, accessed, and changed through a teacher-guided activity using plastic baggies and pieces of paper. The lesson begins with a brief reflection on the value of lists. Students then complete the main activity which introduces the syntax to use lists and the ways they can be used. To wrap up students watch two short videos on lists and record the main concepts in their journals.

Standards

Full Course Alignment

CSP Conceptual Framework

- ▶ **AAP-1** - To find specific solutions to generalizable problems, programmers represent and organize data in multiple ways.
- ▶ **AAP-2** - The way statements are sequenced and combined in a program determines the computed result. Programs incorporate iteration and selection constructs to represent repetition and make decisions to handle varied input values.

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming

Agenda

The Pre-Unit Pulse

Warm Up (5 minutes)

Preview Lists

Activity (30 minutes)

Lists

Wrap Up (10 minutes)

Video and Vocabulary

Assessment: Check For Understanding

Objectives

Students will be able to:

- Use an index to reference specific elements in a list
- Use appropriate vocabulary to describe lists.

Preparation

- 7+ sandwich baggies per pair of students
- 1 gallon-sized baggy per pair of students
- packs of red and orange stickies
- pens / pencils
- 1 dry erase marker per four students (pairs can share)
- Review the slides and click through all animations
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson Modifications**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Unit 6 - Lesson 1 - Lists Explore** - Slides [▼ Make a Copy](#)
- **Lists, Loops, and Traversals** - Slides [▼ Make a Copy](#)

For the students

- [Introduction to Lists - Part 1](#) - Video ([Download](#))
- [Introduction to Lists - Part 2](#) - Video ([Download](#))
- [Unit 6 Journal \(Digital\)](#) - Resource
▼ Make a Copy
- [Unit 6 Journal \(Interactive Notebook\)](#) - Resource
▼ Make a Copy

Vocabulary

- **Data Abstraction** - manage complexity in programs by giving a collection of data a name without referencing the specific details of the representation
- **Element** - an individual value in a list that is assigned a unique index
- **Index** - a common method for referencing the elements in a list or string using numbers
- **List** - an ordered collection of elements

Teaching Guide

The Pre-Unit Pulse

Have your students complete these questions independently some time before starting this unit.

☰ 1-3

The Pre-Unit Pulse

1

2

3

💡 Teaching Tip

Pre-Unit Pulse Questions: The answers to these questions can provide insights into the preferences, strengths, and motivations of your students. If you'd like to adapt the questions or add in your own, you may want to make a copy of [this Google Form](#) which already has the existing questions populated.

For more tips and ideas of how to use these questions, check out the [CSP Guide to Pre-Unit Pulse Questions](#)

Warm Up (5 minutes)

Distribute: Give each student a copy of the Unit 6 Journal ([Digital/Interactive Notebook](#)) or have students set up this unit's section in their CSP journals.

Journaling: Journaling can take many different forms, but in general, it's a tool for individual processing and reflection in a form that can be revisited as students develop their skills and understandings. The medium used for journaling can vary depending on classroom needs.

For more guidance and examples, check out the [CSP Guide to Journaling](#).

Preview Lists

Discuss: *With a partner, brainstorm lists of information that you encounter on a daily basis. Why are lists useful?*

Have students discuss with a partner and then have a few students share out their responses.

Discussion Goal:

- Lists help us group together like information
- We can go through a list item by item to check off things we have completed
- Lists help us stay organized

Activity (30 minutes)

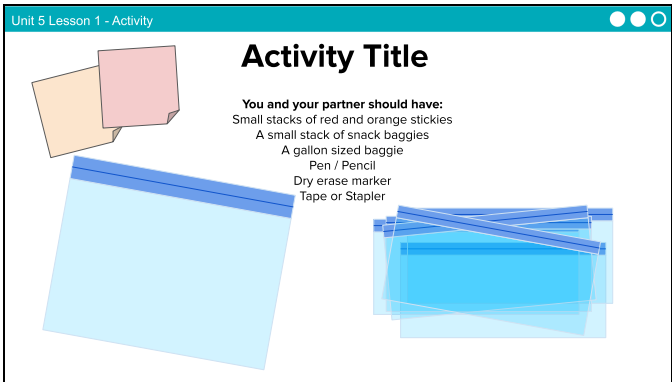
Lists

Group: Group students in pairs.

Distribute: Give each pair of students:

- A small stack of red and orange sticky notes
- A pen/pencil
- 7+ snack-sized plastic baggies
- One gallon-sized plastic baggy
- A dry erase marker to share with another group

Display: Use the activity slides for this lesson to guide the unplugged activity on Conditionals.

Slides	Speaker Notes
	Say: Today, we are going to explore lists. Note: Supplies Substitutions • There's no need to use stickie notes if you have other scraps of colored paper. Also consider cutting stickies in 4 to make them go further. If you don't have dry erase markers handy consider using pieces of masking tape on the baggies. • The gallon-sized baggie can be replaced with any baggie that is large enough to hold the snack-sized baggies.

Slides

Speaker Notes

Unit 5 Lesson 1 - Activity



```
var myName = "José";
```

```
var myName = "José";
```

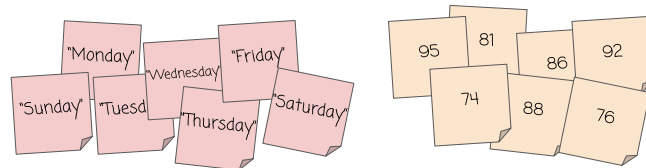
Say: We've already learned that variables can help us store one piece of information.

Unit 5 Lesson 1 - Activity

But what if we have a lot of information?

Do This:

With a partner, discuss the challenges that you'd encounter if you need to store 10, or 100, or 1,000 pieces of information using variables.



Say: But what if we have lots of information?

Do This: With a partner, discuss the challenges that you'd encounter if you need to store 10, or 100, or 1,000 pieces of information using variables.

Discuss: Lead a short discussion on the challenges they'd encounter. The main challenges are that you'd need to make a variable for each piece of information and it could get difficult to name them all.

Unit 5 Lesson 1 - Activity

Creating and Accessing Lists

Say: Let's look at how to use lists

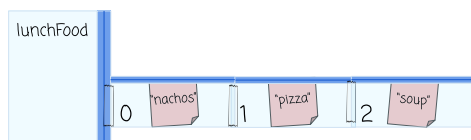
Unit 5 Lesson 1 - Activity

Lists

- Gallon baggie connected to smaller baggies
- Each smaller baggie can hold one value
- Gallon baggies are named with the same rules as a variable (no spaces, can't start with number)
- Smaller baggies are numbered starting at 0

Do This:

Make a list like the one below that holds between 2 and 4 values. Share it with another group.



Say: Today we're going to practice using a new way to store information called a list. Follow the instructions here to make a list of your own.

Do This: Have students create their own list. Circulate the room to check on how they're doing. Make sure they share with another group.

Slides

Speaker Notes

Unit 5 Lesson 1 - Activity

Your list is made up of **elements**. Each element has its own **index**. Indexes start at 0 and count up. The **length** of the list is how many elements it contains. This list has 3 elements and indexes from 0 to 2.

Say: Your list is made up of elements. Each element has its own index. Indexes are just numbers that count up from zero. The length of the list is how many elements it contains. This list has 3 elements and indexes from 0 to 2.

Unit 5 Lesson 1 - Activity

Notice that all the bags can folded up and be placed inside the big variable baggie. Sometimes we want to think about the whole list, sometimes we want to think about individual elements

Say: Notice that all the bags can folded up and be placed inside the big variable baggie. Sometimes we want to think about the whole list, sometimes we want to think about individual elements

Unit 5 Lesson 1 - Activity

var myList = ["pizza", 4]

- This command creates a new list and assigns it to a variable
- A list is indicated with square brackets
- Each value in the list is separated by commas

Do This: Write out the command that creates the list you just made. Then write the command that makes another group's list. Share both answers with that group.

Say: We create a list with this command. It creates a new list and assigns it to a variable. The square brackets indicate we're making a list. Each value is separated by commas.

Do This: Have students write the command to make their own list and the list of partner group. By sharing their answers with another group they can check that they're following along.

Unit 5 Lesson 1 - Activity

myList[1]

- This expression "accesses" the value at that index of the list.
- Also uses square brackets

Do This: What do lunchFood[0] and lunchFood[2] access? Compare your answers with another group.

Say: If we want to access the values in our list we use the square brackets next to the name of our list.

Do This: Have students discuss with other groups. They should see the answers are "nachos" and "soup"

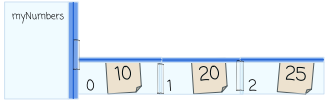
Slides

Speaker Notes

Unit 5 Lesson 1 - Activity

Lists and Expressions

Replace the list access with a copy of the value it holds
Evaluate the expression as normal



3 + myNumbers[1]
3 + 20 evaluates to 23

myNumbers[0] + myNumbers[2]
25 + 10 evaluates to 35

Do This: Write three expression that include accesses to the list you created. Use the examples above for inspiration
Trade with another group and have them evaluate your expressions

Say: We can use list accesses inside of expressions, just like variables.

 Click for animation

Say: This first expression evaluates to 23 because myNumbers[1] contains 20.

 Click for animation

Say: This second expression evaluates to 35 because myNumbers[0] and myNumbers[2] contain 25 and 10.

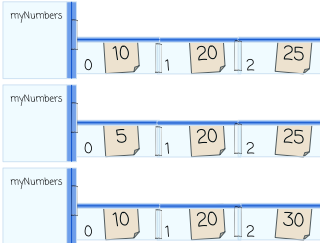
Do This: Have students write three expressions using their own lists. Then have them trade with another group to practice evaluating. Circulate the room to make sure students are following the directions correctly.

Unit 5 Lesson 1 - Activity

myList[1] = "hello"

Assigns the value on the right to the index
Just like variable assignment, the old value is thrown away and replaced.

```
00 var myNumbers = [10,20,25]
01 myNumbers[0] = 5
02 myNumbers[2] = 30
```



Do This: Discuss what the list will contain after line 02 runs.

Say: We can assign the index of a list just like a variable.

Do This: Have students discuss what the list will contain after line 02 runs.

 Click for animation

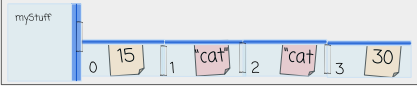
Say: This code will assign a new value at index 2 of the list.

Unit 5 Lesson 1 - Activity

Do This:

Run this program. Compare your result with another group.

```
00 var myStuff = [20, "hat", "pow", 5]
01 myStuff[1] = "cat"
02 myStuff[2] = myStuff[1]
03 myStuff[0] = myStuff[3] + 10
04 myStuff[3] = myStuff[0] + myStuff[0]
```



Do This: Circulate the room while students run this program

 Click for animation

Discuss: Discuss quickly with the class why the program ends with the list shown.

Slides

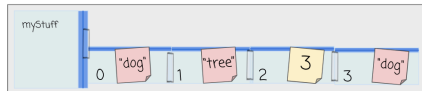
Speaker Notes

Unit 5 Lesson 1 - Activity

Do This:

You can use expressions in the place of the list index.
Run this program and compare your result with another group.

```
00 var myStuff = ["dog", "cat", 3, 10];
01 myStuff[2-1] = "tree";
02 myStuff[myStuff[2]] = myStuff[0];
```



Do This: Circulate the room while students run this program

Click for animation

Discuss: Discuss quickly with the class why the program ends with the list shown.

Note: Line 02 is tricky. You need to first evaluate myStuff[0] which is "dog" to determine the value being assigned. Then evaluate myStuff[2], which is 3. At this point, the command reads myStuff[3] = "dog". Now it is just a simple assignment.

Unit 5 Lesson 1 - Activity

Changing Your List

Say: Now let's learn about three different commands that can change your list.

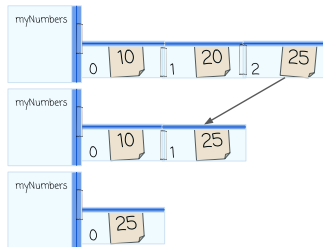
Unit 5 Lesson 1 - Activity

removeItem(list, item)

Removes the item in the list given and at the index given
All items to the right are shifted over
The last index is removed from the list

```
00 var myNumbers = [10, 20, 25]
01 removeItem(myNumbers, 1)
02 removeItem(myNumbers, 0)
```

Do This: Discuss what the list will contain after line 02 runs.



Say: The removeItem command will remove an item from a list. The item at the index given is removed, items to the right are shifted over, and the last index is removed.

Do This: Have students discuss what the list will show after this command.

Click for animation

Discuss: Discuss with the class any questions.

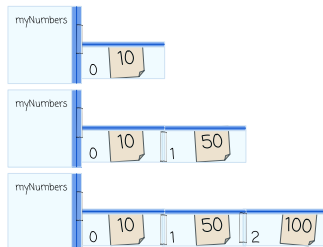
Unit 5 Lesson 1 - Activity

appendItem(list, item)

Adds an item to the end of the list
A new index is added to the list to create a place for the item
The last index is removed from the list

```
00 var myNumbers = [10]
01 appendItem(myNumbers, 50)
02 appendItem(myNumbers, 100)
```

Do This: Discuss what the list will contain after line 02 runs.



Say: The appendItem command will add an item to a list. A new index is added to the end of the list and the new item is place in it.

Do This: Have students discuss what the list will show after this command.

Click for animation

Discuss: Discuss with the class any questions.

Slides

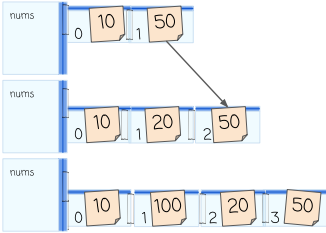
Unit 5 Lesson 1 - Activity

insertItem(list, index, item)

Inserts an item into a list at the index given
A new index is added to the list so there's space for the new item
The new item is placed at the index given, all other items move right

```
00 var nums = [10,50]
01 insertItem(nums,1,20)
02 insertItem(nums,1,100)
```

Do This: Discuss what the list will contain after line 02 runs.



The diagram illustrates the insertion process. It shows three states of a list named 'nums'. Initially, the list contains [10, 50] with indices 0 and 1. After the first insertion, it becomes [10, 20, 50] with indices 0, 1, and 2. After the second insertion, it becomes [10, 100, 20, 50] with indices 0, 1, 2, and 3. Arrows indicate the shifting of elements to the right to make space for the new item.

Speaker Notes

Say: The insertItem command will add an item to the middle of a list. The item is placed in the index given. Then every item is moved over one space to make room. A new index is added to the end of the list to make room.

Do This: Have students discuss what the list will show after this command.

 [Click for animation](#)

Discuss: Discuss with the class any questions.

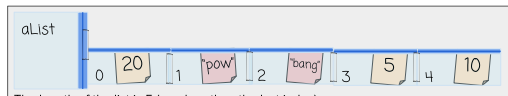
Unit 5 Lesson 1 - Activity

Do This:

Run this program. Compare your result with another group.
What is the length of the list at the end?

```
00 var alist = [20, "hat", "pow"]
01 appendItem(alist, 5)
02 appendItem(alist, 10)
03 removeItem(alist, 1)
04 insertItem(alist, 2, "bang")
```

Command Reference
removeItem(list, index)
appendItem(list, item)
insertItem(list, index, item)



The length of the list is 5 (one less than the last index)

Do This: Have students run this program. Circulate the room to support them with any questions.

 [Click for animation](#)

Discuss: Discuss with the class any questions.

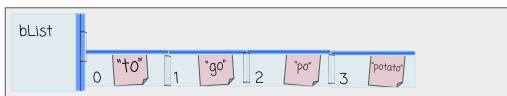
Unit 5 Lesson 1 - Activity

Do This:

Run this program. Compare your result with another group.
What is the length of the list at the end?

```
00 var bList = ["to", 5, "po"]
01 appendItem(bList, bList[2] + "ta" + bList[0])
02 insertItem(bList, 2-1, "go")
03 deleteItem(bList, 2)
```

Command Reference
removeItem(list, index)
appendItem(list, item)
insertItem(list, index, item)



The length of the list is 4 (one less than the last index)

Do This: Have students run this program. Circulate the room to support them with any questions.

 [Click for animation](#)

Discuss: Discuss with the class any questions.

Wrap Up (10 minutes)

Video and Vocabulary

Video: As a class watch both videos on lists.

-  [Introduction to Lists - Part 1](#)
-  [Introduction to Lists - Part 2](#)

 **Do This:** Review Key Takeaways and have students add to their journals: lists, element, and index.

Assessment: Check For Understanding

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: Match the index with the value. `var myNumbers = [32, 64, 33, 0, 15, 26, 3]`



Check For Understanding



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